

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/354219392>

Machine Learning Enabled Fall Detection with Compact Millimeter Wave System

Conference Paper · August 2021

DOI: 10.1109/NAECON49338.2021.9696389

CITATIONS

0

READS

196

4 authors, including:



Abdullah K Alhazmi
University of Dayton

6 PUBLICATIONS 9 CITATIONS

[SEE PROFILE](#)



Mubarak Alayyat Alanazi
University of Dayton

3 PUBLICATIONS 7 CITATIONS

[SEE PROFILE](#)

Machine Learning Enabled Fall Detection with Compact Millimeter Wave System

Abdullah K. Alhazmi
Department of Electrical and
Computer Engineering
University of Dayton
Dayton OH, USA
alhazmia3@udayton.edu

Mubarak A. Alanazi
Department of Electrical and
Computer Engineering
University of Dayton
Dayton OH, USA
alanazim23@udayton.edu

Chengkun Liu
Department of Electrical and
Computer Engineering
University of Dayton
Dayton OH, USA
liuc30@udayton.edu

Vamsy P. Chodavarapu
Department of Electrical and
Computer Engineering
University of Dayton
Dayton OH, USA
vchodavarapu1@udayton.edu

Abstract—Accidental falls have created an alarming situation for elderly people who are either living independently or left alone for a short time such as during a pandemic by their families. Falls have become one of the leading causes of death and irreversible injuries for aged people worldwide. Our proposed system combines machine learning and object detection with low power sensors and computing systems to create a portable and low-cost solution for fall detection. Specifically, our system for human fall detection uses a millimeter wave (mmWave) sensor connected to NVIDIA Jetson Nano system. We use neural network models to analyze the output response from the mmWave system to classify different human postures and provide an accurate fall detection. The developed system provides privacy as no camera images are recorded and high accuracy to detect real-time fall detection for both single and multiple individuals, simultaneously.

Index Terms—Human pose detection, Millimeter wave sensing, Neural network, Nvidia Jetson nano.

I. INTRODUCTION

Incidents related to falls are common among elderly people of age 65 years and more [1]. Falls in older people can cause serious medical conditions such as fractures, severe pain, loss of mobility, and other long-term life-threatening health consequences [2]. Here, smart technologies can help to track and detect human physical and behavioral changes to prevent fall situations by rapidly providing appropriate prompting alerts. Driven by the need to detect and attend to elderly falls in a timely manner, it has become an important area of research and development [3]. To this end, many systems are now commercially available including inertial sensors embedded in jewelry [4], smart watches [5], and emergency contact pendants [6]. Further, facilities offering senior living, independent living, memory care have been increasingly utilizing some of the above-mentioned systems for continuous monitoring of health and wellbeing of people

living in those facilities [7]. However, most of the above mentioned systems rely on people wearing them continuously. This creates limitations of use in many situations such as in bathrooms and bedrooms due to privacy reasons where most falls occur or if the person becomes unconscious due to fall which prevents them from pressing the emergency call button [8].

Radar sensors are non-contact sensors and have been used previously for remote detection of physical human behavior such as pose, position, and presence [9]. Radar sensors can be integrated as part of a smart home monitoring system with rapid response to fall detection and can provide early warning which will dramatically improve elderly care management. Compared to cameras and wearable based systems, radar sensors are less accurate but offer better privacy and convenience of use. Further, standard color-depth cameras (RGB-D) that measure velocity indirectly by translating two successive frames [10], are not reliable in situations where the person loses consciousness while being in a still position.

In prior related research works, Feng *et al.* [11] used three-layer Convolution Neural Network (CNN) model implemented on an embedded graphics processing unit (GPU) board and a millimeter Wave (mmWave) radar hardware platform for fall detection. The model was based on the robot operating system (ROS), which created three nodes on the Nvidia Jetson TX2 platform. A density-based spatial clustering method was used to separate scattering points into multiple targets. The model achieved good performance, however, it used extensive GPU resources, which would limit its implementation on portable embedded systems.

The mmWave sensing technology can be utilized for accurate detection of objects as well as providing the range, velocity, and approaching angle of those objects [12]. The operation range of this contactless technology is in the mil-

limeter wavelength range (frequency range between 30 GHz - 300 GHz). Additionally, it can penetrate certain materials such as plastics and clothing, as well as being impervious to environmental conditions such as rain, fog, and dust. As a result, these systems can provide robust high-resolution signals for fall detection under various circumstances. They are characterized by the following: i) capacity to have large bandwidths which allows to distinguish between various types nearby objects, ii) can be focused and steered which allows large area coverage, and iii) they require relatively small system components such as the antennas and computation.

In our previous work [13], we investigated three Machine Learning (ML) methods to analyze and classify mmWave sensor outputs using a Personal Computer (PC) with MATLAB to predict human poses based on position and distance change. After receiving the signals from the sensor, the ML models classified the input coordinates into one of three positions: standing, sitting, or laying. In this work, we aim to detect human falls and pose by using Neural Networks (NN) implemented on a small, low-power single board computer (SBC), NVIDIA Jetson Nano, to analyze and classify the response from the mmWave sensor (Texas Instruments (TI) IWR6843). The Jetson Nano is a small (69.6 x 45 mm) powerful SBC intended for embedded applications. The Jetson Nano integrates both Graphical Processing Unit (Maxwell architecture with 128 NVIDIA CUDA cores) and Central Processing Unit (Quad-core ARM Cortex-A57 processor) on a single chip. It can run modern ML workloads while consuming less than 5W of power. The remainder of the paper is organized as follows: Section II reviews the related work on mmWave based fall detection, Section III presents a system overview including: mmWave sensor technology, NVIDIA Jetson Nano, and the detection process. The experiment setup and results are presented in Section IV. Finally, the conclusion is presented in the Section V.

II. MILLIMETER WAVE (MMWAVE) BASED FALL DETECTION

mmWave is a type of radar technology that relies on electromagnetic waves and capable of detecting objects with high precision [12]. Many research groups proposed the use of radar technology for fall detection including Moeness *et al.* [14] who utilized Doppler radar technology for fall detection and elderly movement monitoring with high accuracy by using time-frequency characterization. Subsequently, Branka *et al.* [15] deployed a deep learning technique for fall detection using automatic capture of time frequency and range domains. Tao *et al.* [16] used millimeter wave frequency modulated continuous wave (FMCW) radar to analyze the processes related to elderly falls. Feng *et al.* [17] proposed a fall detection system that captured the human body's point cloud along with the body centroid through mmWave radar sensor. Sengupta *et al.* [18] developed mm-Pose that offers human skeleton posture real-time detection. That technique detected over 15 distinct skeletal joints while using mmWave radar reflection signals.

Carlos *et al.* [19] established a robot assisted system with considerably reduced number of parameters for fall detection in real-time scenarios while using NVIDIA Jetson TX2 platform. They deployed pre-trained CNN models for extraction of temporal features through optical flow extraction with a high accuracy of 88.55%. Shengchao *et al.* [20] proposed early fall detection design that is operational on NVIDIA Titan X. The model described achieved a rapid fall detection time of 0.312 seconds for the first fall (average of 0.5 seconds) with 100% accuracy. Ronak *et al.* [21] proposed a fall detection system named as 3D ConvNet architecture using NVIDIA DGX-1 through developing a compressive sensing framework. Tsung *et al.* [22] implemented a fall detection solution with NVIDIA Jetson TX2, for real-time information extraction incorporated with traditional algorithms. The work focused on computation and consumption energy to propose a portable solution with real-time processing capabilities. In contrast to previous works, for the first-time, our proposed ML model is implemented on the Nvidia Jetson nano-board which has the lowest system size, weight, power, and cost (SWaP-C) requirements than prior demonstrated systems.

III. SYSTEM OVERVIEW

A. The Long-Short Term Memory (LSTM) model

The NVIDIA Jetson Nano supports integration with high performance sensors and can process outputs from multiple sensors simultaneously. Our system utilizes the Long-Short Term Memory (LSTM) neural network architecture which employs sequential inputs and maps them to the corresponding output labels. The LSTM model uses consecutive inputs and outputs (indexed by time) to adjust the weights of the networks. Thus, the LSTM creates a memory that improves the performance for time series data based on the constructed temporal relationships [23], [24]. The input training data vector is $X \in \mathbb{R}^n$, where n is the number of training data points, and output vector is $Y \in \mathbb{R}^p$, where $p = 3$ to represent the three output classes. Fig. 1 presents an overview of data training using the LSTM model structure. The LSTM main property is to limit the number of time steps used during network training without losing the memory created by the sequential data, using a cell state, C_t and updating it at every time step, t . The C_t is composed of three main components: the input gate, the forget gate, and the output gate [25].

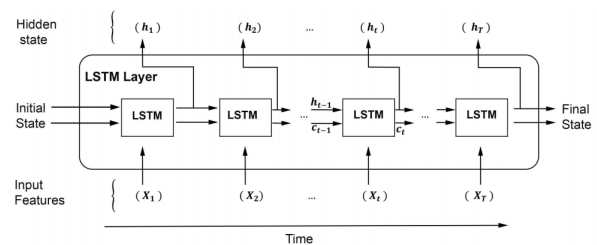


Fig. 1: LSTM model structure

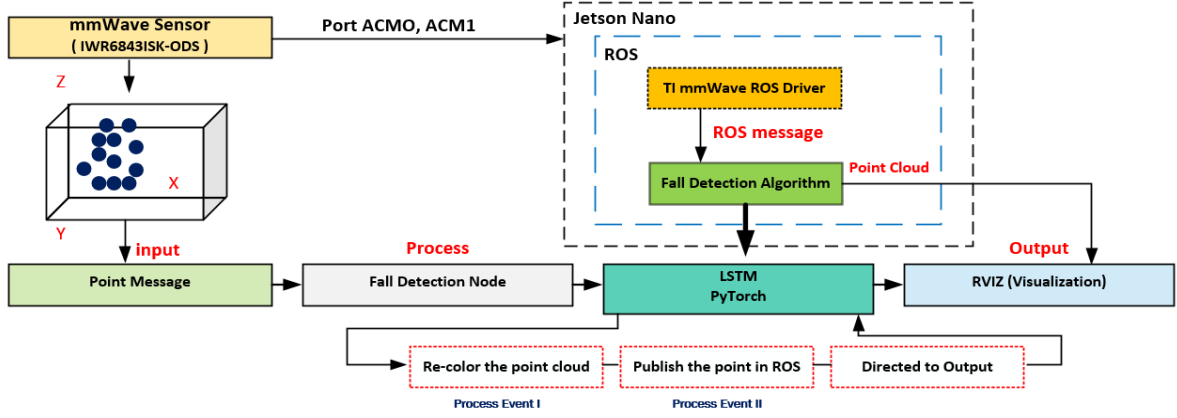


Fig. 2: Detection Process Model and System Timeline.

For every time step t , the forget gate f_t is responsible for deciding the amount of memory (number of time steps) used in the calculations and the amount of data to discard (forget)

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (1)$$

where W is the adjustable weights matrix, h is the hidden state, b is the bias, and σ is the softmax function which outputs a value from 0 (forget everything) to 1 (keep everything). Next, the cell state decides the amount of memory to be stored in C_t as follows:

$$C_t = f_t * C_{t-1} + i_t * C'_t \quad (2)$$

where the first term is the update of the old cell state C_{t-1} multiplied by f_t , and the second term is the multiplication of the candidate values C'_t and the input gate i_t . The input gate is responsible for deciding which values to be updated, and is formulated as:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \quad (3)$$

The candidate values C'_t used in 2 are generated by

$$C'_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (4)$$

where the $\tanh(\cdot)$ function is an activation function that outputs values between -1 and 1. The final step is generating the output by using the output function o_t

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o). \quad (5)$$

The LSTM model is convenient to classify, process, and predict events in time series data actively for scenarios that have unexpected and unknown duration [26]. This model is advantageous for our proposed system as it requires 10% less training time and possesses low energy and space requirements for its real-time operations [27]. LSTM is further favorable due to being insensitive to indefinite time gaps, which fulfills an essential and integral requirement of our system.

B. System Parameters

Parameters used in the experimental analysis are reported in Table I. The fall detection node has been developed as a Robot Operating System (ROS) package and was implemented in Python. It was designed to communicate with a mmWave ROS driver based on topic/message communication. We integrated a LSTM algorithm into the fall detection software based on the PyTorch library to detect the human conditions [11]- [17]. The LSTM algorithm consists of two fully connected networks interconnected with each other. Moreover, our hardware includes one mmWave sensor along with a Jetson Nano connected to a PC, which are connected to each other through the ttyACM port. We used the ROS middle-ware to connect the sensor to our software in order to facilitate the communication and attain seamless communications. All the point clouds captured by mmWave sensor and ported to the mmWave ROS driver are provided by TI and published as a ROS environment message.

TABLE I: Experiment parameters

Parameters	Physical Description
Frequency Band	60-64 GHz
Platform	IWR68xx
Operating System	ROS (Linux Machine)
Visualization	Radar pointcloud using Rviz
Port	ttyACM
Frame Rate	10 fps
Range Resolution	0.044 m
Maximum unambiguous Range	9.02 m
Maximum Radial Velocity	1 m/s
Radial Velocity Resolution	0.13 m/s
Range Detection Threshold	15 dB
Doppler Detection Threshold	15 dB

The detection process model and system overview are shown in Fig. 2 with a detailed flow of a structured timeline. The system perspective is composed of two interconnected processes executing side by side, which are the detection process and detection pipeline. The two interconnected processes are explained in the two following subsections.

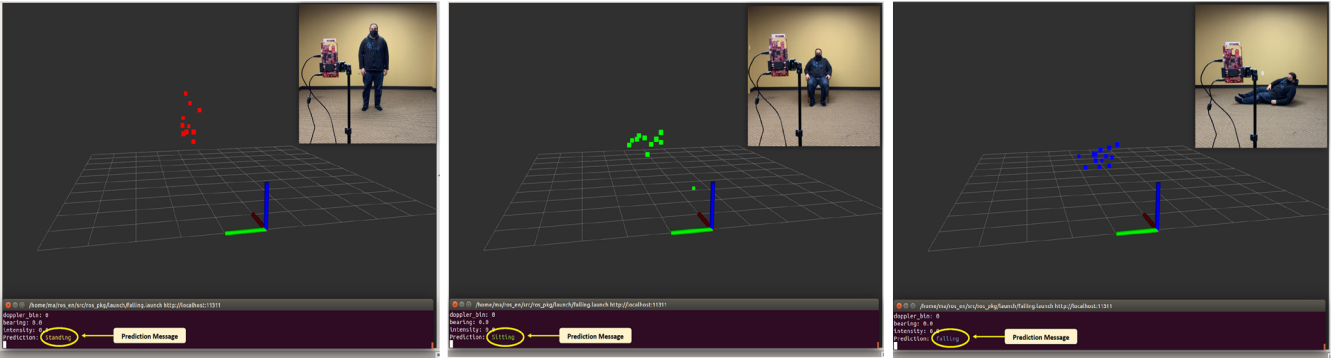


Fig. 3: Different subject positions with corresponding outputs including standing (left), sitting (middle), and falling (right).

C. Detection Process

First, the detection process is initiated when receiving input (point message), and the input is pre-processed according to respective change in coordinates and by detecting node. Coordinates i.e. x, y, z are explicitly designed to act as data acquisition points. Afterwards, the fall detection node starts the process by triggering the input through LSTM which is capable of learning order dependence in the sequence efficiently. The LSTM robustly identifies machine translation behavior even in complex scenarios. Both the bidirectional coordination and sequence-to-sequence communication pipeline are well-supported in this model. Upon the LSTM activation (Process Event I), a specialized color rich point cloud shows response against a message, i.e. standing is indicated by red color, sitting is indicated by green color and falling (laying down) is indicated by blue color. By identifying the color indicators, Process Event II publishes the point to the ROS and leads to the directed output known as RVIZ. The integrated visualizer window displays all point clouds in the ROS color coordinated messages.

D. Detection Pipeline

TI mmWave sensor is functional on ports ACMO and ACM1. When the mmWave sensor transposes the coordinates (collected data of x, y, z coordinates), the TI mmWave Robot Operating System (ROS) driver performs an operation with the LSTM fall detection algorithm via ROS message and the specific formatted message is diverted to RVIZ. The LSTM is capable of recognizing the sequences of images. The mmWave ROS driver is reliable and compatible to bridge ROS operations and framework to facilitate a Radar point cloud and enables smooth execution of a point cloud reaching RVIZ. The ROS is integrated with NVIDIA Jetson Nano that effectively assists in detecting the pipeline response rate, accuracy, and results in less error ratios.

IV. EXPERIMENT RESULTS AND DISCUSSION

A. Experimental Arrangements

The experimental setup was placed in a large room to assure proper testing of the sensor functionality. Fig. 4 illustrates the detailed experimental arrangement, where the human is the

target to be detected (the person is seen wearing face mask due to COVID-19 pandemic regulations), NVIDIA Jetson Nano Board is connected to both the TI IWR6843 mmWave Sensor and a PC.

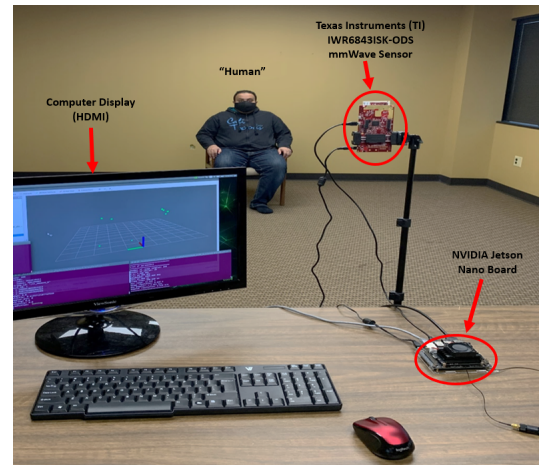


Fig. 4: The experimental setup.

In the experiment, three different positions (i.e., laying down, standing, and sitting) are considered as the three output classes. The positions coordinates related to the target posture are outlined with different colors as shown in Fig. 3. The three distinct colors (red, green, and blue) assist in quick visualization of different postures (standing, sitting, and laying down) without the need to interpret the coordinates. The mmWave sensor is capable of detecting multiple people concurrently. This feature is vital in order to detect falls when more than one person is present in its field of view. Fig. 5 portrays three targets with three different positions simultaneously.

B. Data Collection and Training

In order to produce sufficient training data, the experiment was repeated 20 times with various number of participants and different sensor placements. In each experiment trial, the data points of a random number of participants between one and seven were recorded. A wide range of human and height combinations were considered during the experiment to create

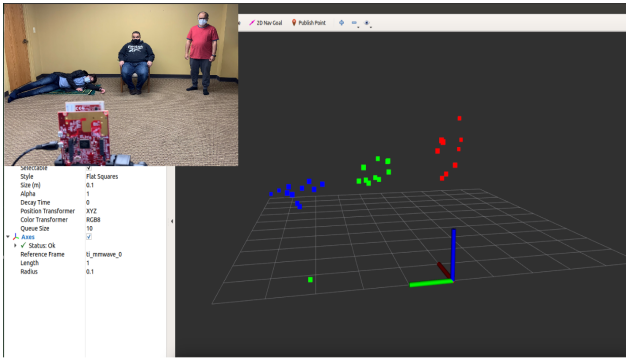


Fig. 5: Simultaneous position detection by the mmWave sensor for three different persons.

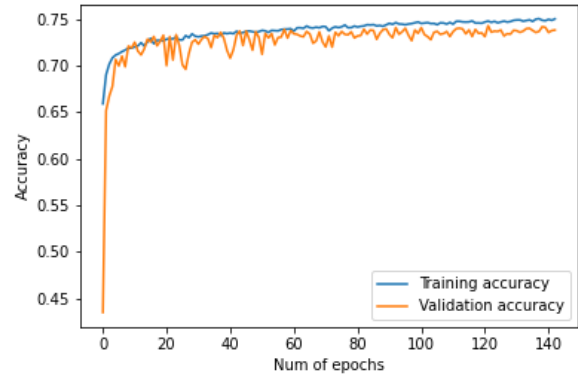
a representative training data. The variance in the weight height parameters during the training phase is essential for the model to have a sufficient representative data points such that it can generalize its classification ability using the test data. The data was manually labelled for each position during the data collection phase.

C. Model Performance Analysis

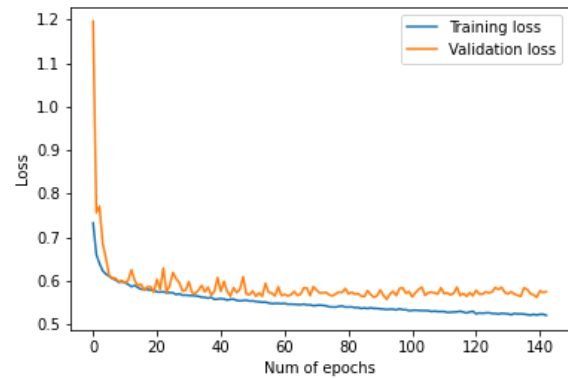
For the performance analysis, we split the collected data into training (75%) and validation data (25%). The model is trained on the training data, and then the performance is evaluated on the previously unseen validation data. The training and validation accuracy during the training phase are depicted in Fig. 6(a), while the training and validation loss are shown in Fig. 6(b). As shown in the two graphs, the increase in the accuracy and the decrease in loss function indicate that the model is adjusting the weights to correctly classify the three positions. Furthermore, the improvement of accuracy and loss function for the validation with the epochs progress indicate that the model can generalize and accurately predict the outcome on validation data.

In addition to the accuracy and loss function metric, we use the Receiver Operating Curve (ROC) to assess the performance of the model. The ROC diagnoses the classification ability by depicting the true positive rate against false positive rate for various discrimination thresholds. The ROC for the three designated classes confirms the superior performance of the proposed model for all the classes of motion as illustrated in Fig. 7. It is shown that the ROC for the classes indicates worthwhile peaks with true incidence occurrences performance for fall detection scenarios as well. ROC curves for all three classes are parallel to each other in performance which indicates its better true positioning rate equally for all designated postures.

The sequence of fall detection is depicted in Fig. 8 in four steps, from Fig. 8(a) to Fig. 8(d). The sequence shows the transition from standing to falling, and finally laying on the ground. The coordinates (with respective colors) are shown next to each position.



(a)



(b)

Fig. 6: The results for training and validation data; (a) accuracy and (b) loss function.

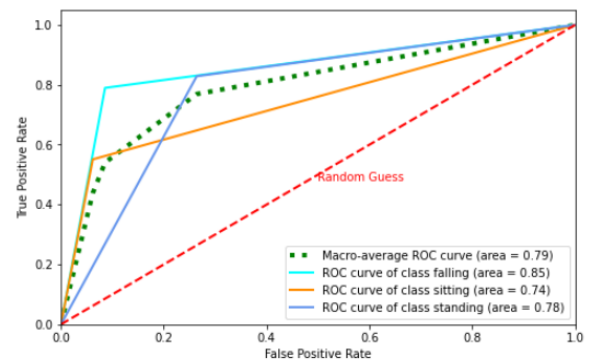


Fig. 7: LSTM model ROC curves for the three classes.

V. CONCLUSION

We described an efficient and prompt fall detection system using a mmWave sensor and a NVIDIA Jetson Nano single board computer implementing ML algorithms to analyze and classify the human pose and position while maintaining the

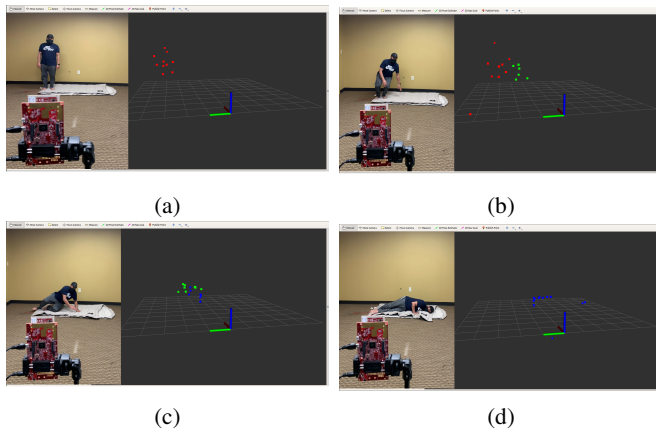


Fig. 8: Real-time human behavior and their corresponding point clouds: From standing (a) to laying down (d).

privacy for the user. The system provided good performance for classifying the input coordinates into one of three positions: standing, sitting, and laying down. The system detected single posture as well as multi-posture occurrences. The proposed system can act as an optimized solution for wellbeing of elderly people who are susceptible to life threatening injuries due to accidental falls.

REFERENCES

- [1] S. Hopewell, B. Copey, P. Nicolson, B. Adedire, G. Boniface, and S. Lamb, "Multifactorial interventions for preventing falls in older people living in the community: a systematic review and meta-analysis of 41 trials and almost 20 000 participants," *British journal of sports medicine*, vol. 54, no. 22, pp. 1340–1350, 2020.
- [2] G. F. Papalia, R. Papalia, L. A. Diaz Balzani, G. Torre, B. Zampogna, S. Vasta, C. Fossati, A. M. Alifano, and V. Denaro, "The effects of physical exercise on balance and prevention of falls in older people: A systematic review and meta-analysis," *Journal of clinical medicine*, vol. 9, no. 8, p. 2595, 2020.
- [3] K. Chaccour, R. Darazi, A. H. El Hassani, and E. Andres, "From fall detection to fall prevention: A generic classification of fall-related systems," *IEEE Sensors Journal*, vol. 17, no. 3, pp. 812–822, 2016.
- [4] J. A. Santoyo-Ramón, E. Casilari, and J. M. Cano-García, "Analysis of a smartphone-based architecture with multiple mobility sensors for fall detection with supervised learning," *Sensors*, vol. 18, no. 4, p. 1155, 2018.
- [5] E. Casilari and M. A. Oviedo-Jiménez, "Automatic fall detection system based on the combined use of a smartphone and a smartwatch," *PloS one*, vol. 10, no. 11, p. e0140929, 2015.
- [6] E. L. Chuma, L. L. B. Roger, G. G. de Oliveira, Y. Iano, and D. Pajuelo, "Internet of things (iot) privacy-protected, fall-detection system for the elderly using the radar sensors and deep learning," in *2020 IEEE International Smart Cities Conference (ISC2)*. IEEE, pp. 1–4.
- [7] L. Yu, W. M. Chan, Y. Zhao, and K.-L. Tsui, "Personalized health monitoring system of elderly wellness at the community level in hong kong," *IEEE Access*, vol. 6, pp. 35 558–35 567, 2018.
- [8] J.-x. Liu, R. Tan, N. Sun, G. Han, and X.-f. Li, "Fall detection under privacy protection using multi-layer compressed sensing," in *2020 3rd International Conference on Artificial Intelligence and Big Data (ICAIBD)*. IEEE, 2020, pp. 247–251.
- [9] L. Liu, M. Popescu, M. Skubic, M. Rantz, T. Yardibi, and P. Cuddihy, "Automatic fall detection based on doppler radar motion signature," in *2011 5th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth) and Workshops*. IEEE, 2011, pp. 222–225.
- [10] B. Jekanovic, M. G. Amin, and F. Ahmad, "Effect of data representations on deep learning in fall detection," in *IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM)*, 2016, pp. 1–5.
- [11] F. Jin, R. Zhang, A. Sengupta, S. Cao, S. Hariri, N. K. Agarwal, and S. K. Agarwal, "Multiple patients behavior detection in real-time using mmwave radar and deep cnns," in *2019 IEEE Radar Conference (RadarConf)*. IEEE, 2019, pp. 1–6.
- [12] C. Iovescu and S. Rao, "The fundamentals of millimeter wave sensors," *Texas Instruments, SPYY005*, 2017.
- [13] M. Alanazi, A. Alhazmi, C. Yakopcic, and V. Chodavarapu, "Machine learning models for human fall detection using millimeter wave sensor," *IEEE 55th Annual Conference on Information Sciences and Systems (CISS)*, 2021.
- [14] M. G. Amin, Y. D. Zhang, F. Ahmad, and K. D. Ho, "Radar signal processing for elderly fall detection: The future for in-home monitoring," *IEEE Signal Processing Magazine*, vol. 33, no. 2, pp. 71–80, 2016.
- [15] B. Jekanović and M. Amin, "Fall detection using deep learning in range-doppler radars," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, no. 1, pp. 180–189, 2018.
- [16] T. Yang, J. Cao, and Y. Guo, "Placement selection of millimeter wave fmcw radar for indoor fall detection," in *IEEE MTT-S International Wireless Symposium (IWS)*, 2018, pp. 1–3.
- [17] F. Jin, A. Sengupta, and S. Cao, "mmfall: Fall detection using 4-d mmwave radar and a hybrid variational rnn autoencoder," *IEEE Transactions on Automation Science and Engineering*, 2020.
- [18] A. Sengupta, F. Jin, R. Zhang, and S. Cao, "Mm-pose: Real-time human skeletal posture estimation using mmwave radars and cnns," *IEEE Sensors Journal*, vol. 20, no. 17, pp. 10 032–10 044, 2020.
- [19] C. Menacho and J. Ordoñez, "Fall detection based on cnn models implemented on a mobile robot," in *IEEE 17th International Conference on Ubiquitous Robots (UR)*, 2020, pp. 284–289.
- [20] S. Li, H. Xiong, and X. Diao, "Pre-impact fall detection using 3d convolutional neural network," in *IEEE 16th International Conference on Rehabilitation Robotics (ICORR)*, 2019, pp. 1173–1178.
- [21] R. Gupta, P. Anand, S. Chaudhury, B. Lall, and S. Singh, "Compressive sensing based privacy for fall detection," in *National Conference on Computer Vision, Pattern Recognition, Image Processing, and Graphics*. Springer, 2019, pp. 429–438.
- [22] T.-H. Tsai and C.-W. Hsu, "Implementation of fall detection system based on 3d skeleton for deep learning technique," *IEEE 8th Global Conference on Consumer Electronics (GCCE)*, vol. 7, pp. 153 049–153 059, 2019.
- [23] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [24] Y. Bengio, I. Goodfellow, and A. Courville, *Deep learning*. MIT press Massachusetts, USA:, 2017, vol. 1.
- [25] K. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "Lstm: A search space odyssey," *IEEE transactions on neural networks and learning systems*, vol. 28, no. 10, pp. 2222–2232, 2016.
- [26] Y. Sun, R. Hang, Z. Li, M. Jin, and K. Xu, "Privacy-preserving fall detection with deep learning on mmwave radar signal," in *IEEE Visual Communications and Image Processing (VCIP)*, 2019, pp. 1–4.
- [27] L. Ma, M. Liu, N. Wang, L. Wang, Y. Yang, and H. Wang, "Room-level fall detection based on ultra-wideband (uwb) monostatic radar and convolutional long short-term memory (lstm)," *Sensors*, vol. 20, no. 4, p. 1105, 2020.